

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ES-201

Subject: Computational Methods

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data if any. Scientific calculator is allowed.

Q1 Attempt all questions:

- a) The height of an observation tower was estimated to be 47m, whereas its actual height was 45m. Calculate the percentage relative error in the measurement. (1.5)
- b) Derive the formula for evaluating $\sqrt{12}$ by Newton's Raphson method. (1.5)
- c) Differentiate between partial and complete pivoting in solving linear system of algebraic equations. (1.5)
- d) Estimate the missing term in the following table: (1.5)
- | | | | | | |
|---------|---|---|---|---|----|
| $x:$ | 0 | 1 | 2 | 3 | 4 |
| $f(x):$ | 1 | 3 | 9 | — | 81 |
- e) Define cubic spline function and state the conditions required for cubic spline interpolation. (1.5)
- f) Find $\int_0^{\pi/2} \sqrt{\cos \theta} d\theta$ using Simpson's 1/3rd rule for $n=6$. 1.187 (2.5)
- g) Solve the IVP $\frac{dy}{dx} = xe^y, y(0)=0$ using Picard's method and estimate $y(0.2)$. (3)
- h) Determine which of the following equations are elliptic, parabolic, and hyperbolic. (2)
- (i) $f_{xx} + 6f_{xy} + 9f_{yy} = 0$
- (ii) $f_{xx} - 2f_{xy} + 2f_{yy} = 2x + 5y$

UNIT-I

- Q2 a) Find the root of the equation $x = e^{-x}$, correct to three decimal places by Secant method by performing six iterations. (8)
- b) Use Newton—Raphson method to obtain a root of $\sin x = 1 - x$ to three decimal places. (7)
- Q3 a) Perform four iterations of Golden section search method to minimize $f(x) = x^4 - 14x^3 + 60x^2 - 70x$, $x \in (0, 2)$ with $\epsilon = 10^{-3}$. (7.5)
- b) Use steepest descent method for 3 iterations on $f(x_1, x_2, x_3) = (x_1 - 4)^4 + (x_2 - 3)^2 + 4(x_3 + 5)^4$ with initial point $x^{(0)} = [4, 2, -1]^T$ (7.5)

UNIT-II

- Q4 a) Prove that $\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{1 + \delta^2/4}$. (7)
- b) The table gives the distance in nautical miles of the visible horizon for the given heights in feet above the earth's surface: (8)

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$x = \text{height} :$	100	150	200	250	300	350	400
$y = \text{distance} :$	10.63	13.03	15.04	16.81	18.42	19.90	21.27

Find the value of y when $x=410$ using Newton's backward interpolation formula.

- Q5 a) Evaluate the integral $\int_0^6 \frac{e^x}{1+x} dx$ using Simpson's $1/3^{\text{rd}}$ and Simpson's $3/8^{\text{th}}$ rule respectively. Compare it with the exact value. (7)
- b) Evaluate the integral $\int_0^{0.5} \left(\frac{x}{\sin x} \right) dx$ using Romberg's method, correct to three decimal places. (8)

UNIT-III

- Q6 a) Investigate the values of λ and μ so that the system of equations (7)
- $$\begin{aligned} 2x+3y+5z &= 9 \\ 7x+3y-2z &= 8 \\ 2x+3y+\lambda z &= \mu \end{aligned}$$

have (i) no solution, (ii) unique solution, (iii) an infinite number of solutions.

- b) Solve the system of equations (8)

$$\begin{cases} x_1 + 10x_2 - x_3 = 3 \\ 2x_1 + 3x_2 + 20x_3 = 7 \\ 10x_1 - x_2 + 2x_3 = 4 \end{cases}$$

Using the Gauss elimination with partial pivoting.

- Q7 a) Solve the system of equations using Dolittle factorisation method (9)

$$\begin{cases} 3x+2y+7z=4 \\ 2x+3y+z=5 \\ 3x+4y+z=7 \end{cases}$$

- b) Determine the numerically dominant eigenvalue and eigenvector of

the matrix $A = \begin{bmatrix} 15 & -4 & -3 \\ -10 & 12 & -6 \\ -20 & 4 & -2 \end{bmatrix}$ using Power method. Take the initial

vector $X^{(0)} = [1, 1, 1]^T$.

(6)

UNIT-IV

- Q8 a) Employ Taylor's method to obtain the approximate value of y at $x=0.2$ for the differential equation $dy/dx = 2y + 3e^x$, $y(0)=0$. Compare the numerical solution obtained with the exact solution. (7)

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- b) Using Runge-Kutta method of fourth order, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2, 0.4$. (8)

- Q9 a) Solve the initial value problem $y \frac{dy}{dx} = x$, $y(0) = 1$, using Euler's method in $0 \leq x \leq 0.8$, with $h = 0.2$. Compare the results with the exact solution at $x = 0.8$. (8)

- b) Solve the partial differential equation (7)

$$u_{xx} + u_{yy} = x + y + 1, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1,$$

$$u = 0 \text{ on the boundary}$$

Numerically up to three iterations with $h = 1/3$. Obtain the results correct to three decimal places.
